

KENDRIYA VIDYALAYA SANGATHAN, GURUGRAM REGION

(WINTER STATION)

PRE-BOARD EXAMINATION (2024-25)

Class: XII

Max Marks: 70

Subject: Physics (Theory)

Time : 3 hrs

General Instructions:

- (1) All questions are compulsory. There are 33 questions in all.
- (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- (3) Section A contains twelve very short answer questions and four assertion reasoning MCQs of 1 mark each, Section B contains five short answer questions of 2 marks each, Section C contains seven short answer questions of 3 marks each, Section D has two case study based questions of 4 marks each, and Section E contains three long answer questions of 5 marks each.
- (4) There is no overall choice. However internal choice is provided. You have to attempt only one of the choices in such questions.
- (5) Use of calculators is not permitted.
- (6) You may use the following values of physical constants wherever

$$c = 3 \times 10^8 \text{ ms}^{-1}, h = 6.6 \times 10^{-34} \text{ Js}$$

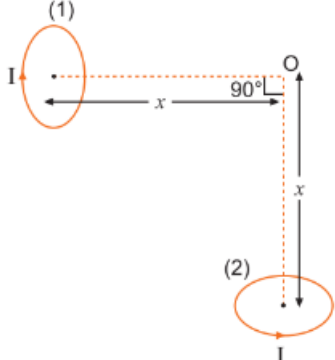
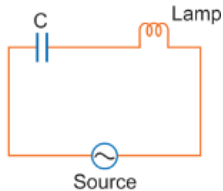
$e = 1.6 \times 10^{-19} \text{ C}$, $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$, Mass of neutron $m_n = 1.6 \times 10^{-27} \text{ kg}$, Boltzmann's constant $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$, Avogadro's number $N_A = 6.023 \times 10^{23} / \text{mole}$

Sr. No.	Section – A	Marks
1	When air is replaced by a medium of dielectric constant K, the force of attraction between two charges separated by a distance r- (a) decreases K times (b) remains unchanged (c) increases K times (d) increases K^{-2} times	1
2	A charge Q is enclosed by a Gaussian spherical surface of radius R. If the radius is doubled, then the outward electric flux will- (a) be doubled (b) increase four times (c) be reduced to half (d) remain the same	1
3	Two identical metal plates, separated by a distance d form a parallel-plate capacitor. A metal sheet of thickness d/2 is inserted between the plates. The ratio	1

	(c) leads the applied emf by an angle $\pi / 2$ (d) and applied emf are in same phase	
10	Out of the following options which one can be used to produce a propagating electromagnetic wave? (a) A chargeless particles (b) An accelerating charge (c) A charge moving at constant velocity (d) A stationary charge	1
11	The stopping potential in an experiment on photoelectric effect is 1.5 V. What is the maximum kinetic energy of the photoelectrons emitted? (a) 2.4×10^{-19} J (b) 2.4×10^{19} J (c) 5.4×10^{-19} J (d) 3.4×10^{-15} J	1
12	The size of the nucleus is proportional to- (a) A (b) $A^{1/3}$ (c) $A^{2/3}$ (d) $A^{-1/3}$	1
Q. No. 13-16 are assertion and reason-based questions having options as given below: - (A) Both Assertion and reason are true and reason is correct explanation of assertion. (B) Both Assertion and reason both are true but reason is not the correct explanation of assertion. (C) Assertion is true, reason is false. (D) Assertion is false, reason is also false.		
13	Assertion: The focal length of the convex mirror will increase, if the mirror is placed in water. Reason: The focal length of a convex mirror of radius R is equal to, $f = 2 R$.	1
14	Assertion: Photoelectric saturation current increases with the increase in frequency of incident light. Reason: Energy of incident photons increases with increase in frequency and as a result photoelectric current increase.	1
15	Assertion: Density of all the nuclei is same. Reason: Radius of nucleus is directly proportional to the cube root of mass number.	1
16	Assertion: A pure semiconductor has negative temperature coefficient of resistance. Reason: In a semiconductor on raising the temperature, more charge carriers are released, conductance increases and resistance decreases.	1
Section – B (2 Marks Questions)		
17	Define electric dipole moment. Is it a scalar or a vector quantity? What are its SI unit?	2

18	Identify the electromagnetic waves whose wavelengths vary as (a) $10^{-12}\text{m} < \lambda < 10^{-8}\text{m}$ (b) $10^{-3}\text{m} < \lambda < 10^{-1}\text{m}$ Write one use for each.	2
19	How will the interference pattern in Young's double slit experiment get affected, when- (i) distance between the slits S_1 and S_2 reduced and (ii) the entire set up is immersed in water? Justify your answer.	2
20	A hydrogen atom in the ground state is excited by an electron beam of 12.5 eV energy. Find out the maximum number of lines emitted by the atom from its excited state.	2
21	Distinguish between a metal and an insulator on the basis of energy band diagrams.	2

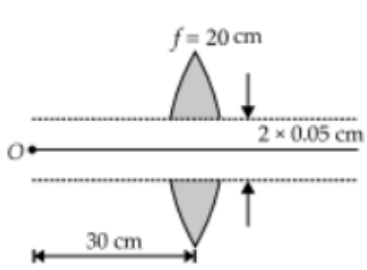
Section – C (3 Marks Questions)

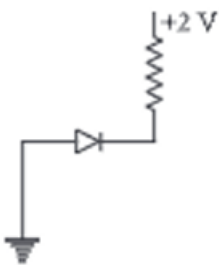
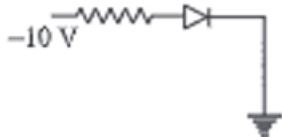
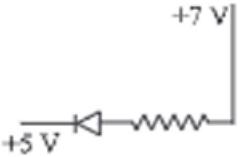
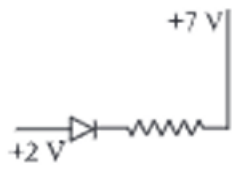
22	Two small identical circular loops, marked (1) and (2), carrying equal currents, are placed with the geometrical axes perpendicular to each other as shown in the figure. Find the magnitude and direction of the net magnetic field produced at the point O.  OR A wire AB is carrying a steady current of 12 A and is lying on the table. Another wire CD carrying 5 A is held directly above AB at a height of 1 mm. Find the mass per unit length of the wire CD so that it remains suspended at its position when left free. Give the direction of the current flowing in CD with respect to that in AB. [Take the value of $g = 10\text{ ms}^{-2}$]	3
23	Define the terms (i) current sensitivity and (ii) voltage sensitivity of a galvanometer. Why does increasing the current sensitivity not necessarily increase voltage sensitivity?	3
24	Define self-inductance of a coil. Show that magnetic energy required to build up the current I in a coil of self-inductance L is given by $\frac{1}{2}LI^2$.	3
25	An electric lamp connected in series with a capacitor and an ac source is glowing with of certain brightness.  How does the brightness of the lamp change on reducing the (i) capacitance and (ii) frequency?	3

26	(a) Why are coherent sources necessary to produce a sustained interference pattern? (b) In Young's double slit experiment using monochromatic light of wavelength λ, the intensity of light at a point on the screen where path difference is λ, is K units. Find out the intensity of light at a point where path difference is $\lambda/3$.	3
27	Using photon picture of light, show how Einstein's photoelectric equation can be established. Write two features of photoelectric effect which cannot be explained by wave theory.	3
28	Draw the graph showing the variation of binding energy per nucleon with the mass number for a large number of nuclei $2 < A < 240$. What are the main inferences from the graph? Explain with the help of this plot the release of energy in the processes of nuclear fission and fusion.	3

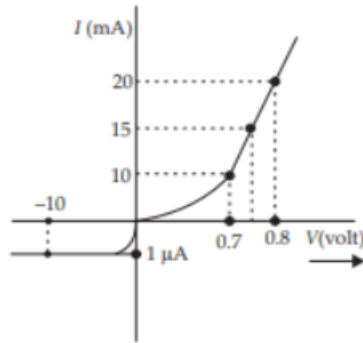
Section – D (case Study)

Attempt any four questions out of five in each case.

29	A convex or converging lens is thicker at the center than at the edges. It converges a parallel beam of light on refraction through it. It has a real focus. A convex lens is of three types: (i) Double convex lens (ii) Plano-convex lens (iii) Concavo-convex lens. The Concave lens or Diverging lens is thinner at the center than at the edges. It diverges a parallel beam of light on refraction through it. It has a virtual focus. (i) A point object O is placed at a distance of 0.3 m from a convex lens (focal length 0.2 m) cut into two halves each of which is displaced by 0.0005 m as shown in the figure. What will be the location of the image?  (a) 30 cm right of lens (b) 60 cm right of lens (c) 70 cm left of lens (d) 40 cm left of lens (ii) Two thin lenses are in contact and the focal length of the combination is 80 cm. If the focal length of one lens is 20 cm, the focal length of the other would be-	4
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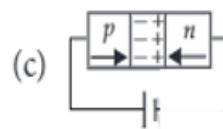
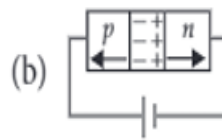
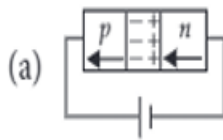
	(a) -26.7 cm (b) 60 cm (c) 80 cm (d) 20 cm (iii) A spherical air bubble is embedded in a piece of glass. For a ray of light passing through the bubble, it behaves like a (a) converging lens (b) diverging lens (c) plano-converging lens (d) plano-diverging lens OR Lens used in magnifying glass is (a) Concave lens (b) Convex lens (c) Both (a) and (b) (d) Neither concave lens nor convex lens (iv) The magnification of an image by a convex lens is positive only when the object is placed (a) at its focus F (b) between F and $2F$ (c) at $2F$ (d) between F and optical centre	
30	When the diode is forward biased, it is found that beyond forward voltage $V = V_k$, called knee voltage, the conductivity is very high. At this value of battery biasing for p-n junction, the potential barrier is overcome and the current increases rapidly with an increase in forwarding voltage. When the diode is reverse biased, the reverse bias voltage produces a very small current about a few microamperes which almost remains constant with bias. This small current is reverse saturation current. (i) In which of the following figures, the p-n diode is forward biased- (a)  (b)  (c)  (d)  (ii) Based on the V-I characteristics of the diode, we can classify diode as (a) bi-directional device (b) ohmic device (c) non-ohmic device (d) passive element (iii) The V-I characteristic of a diode is shown in the figure. The ratio of forward to	4

reverse bias resistance is-



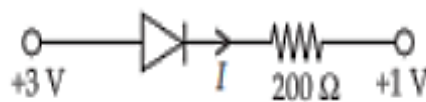
- (a) 100 (b) 10^6
(c) 10 (d) 10^{-6}

(iv) In the case of forwarding biasing of a p-n junction diode, which one of the following figures correctly depicts the direction of conventional current (indicated by an arrow mark)?



OR

(v) If an ideal junction diode is connected as shown, then the value of the current I is-



- (a) 0.013 A (b) 0.02 A
(c) 0.1 A (d) 0.01 A

Section – E

All questions are compulsory. In case of internal choices, attempt anyone.

31

- (a) Define electric flux. Write its SI unit.
(b) Using Gauss's law, prove that the electric field at a point due to a uniformly charged infinite plane sheet is independent of the distance from it.
(c) How is the electric field directed if (i) the sheet is positively charged, (ii) negatively charged?

OR

	Write an expression for the energy stored per unit volume in a charged parallel plate capacitor. How is the energy of parallel plate capacitor changes when a dielectric medium of dielectric constant k is inserted between the plates of capacitor connected with battery? (b) Find the ratio of the potential differences that must be applied across the parallel and series combination of two capacitors C_1 and C_2 with their capacitances in the ratio 1 : 2 so that the energy stored in the two cases becomes the same.	
32	(i) On the basis of electron drift, derive an expression for resistivity of a conductor in terms of number density of free electrons and relaxation time. On what factors does resistivity of a conductor depend? (ii) Why alloys like constantan and manganin are used for making standard resistors? OR Draw a circuit diagram showing balancing of Wheatstone bridge. Use Kirchhoff's rules to obtain the balance condition in terms of the resistances of four arms of Wheatstone Bridge. Two conducting wires X and Y of same diameter but different materials are joined in series across a battery. If the number density of electrons in X is twice that in Y, find the ratio of drift velocity of electrons in the two wires.	5
33	(i) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power. (ii) You are given three lenses of power 0.5 D, 4 D and 10 D to design a telescope. (a) Which lenses should be used as objective and eyepiece? Justify your answer. (b) Why is the aperture of the objective preferred to be large? OR Define the term wavefront. Using Huygen's wave theory, verify the law of reflection. A parallel beam of light of wavelength 600 nm is incident normally on a slit of width 'a'. If the distance between the slit and the screen is 0.8 m and the distance	5

	of 2nd order maximum from the centre of the screen is 1.5 mm, calculate the width of the slit.	
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